Current Figure and Table Reference

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This file collects the current manuscript displays. Recovery is defined as

$$\rho = \frac{\mathrm{acc}(\mathrm{patch}) - \mathrm{acc}(\mathrm{receiver})}{\mathrm{acc}(\mathrm{donor}) - \mathrm{acc}(\mathrm{receiver})}. \tag{1}$$

Source-attention dependence is $d(L) = 1 - q(L)$, the fraction of originally correct answers lost under cumulative source-attention knockout.

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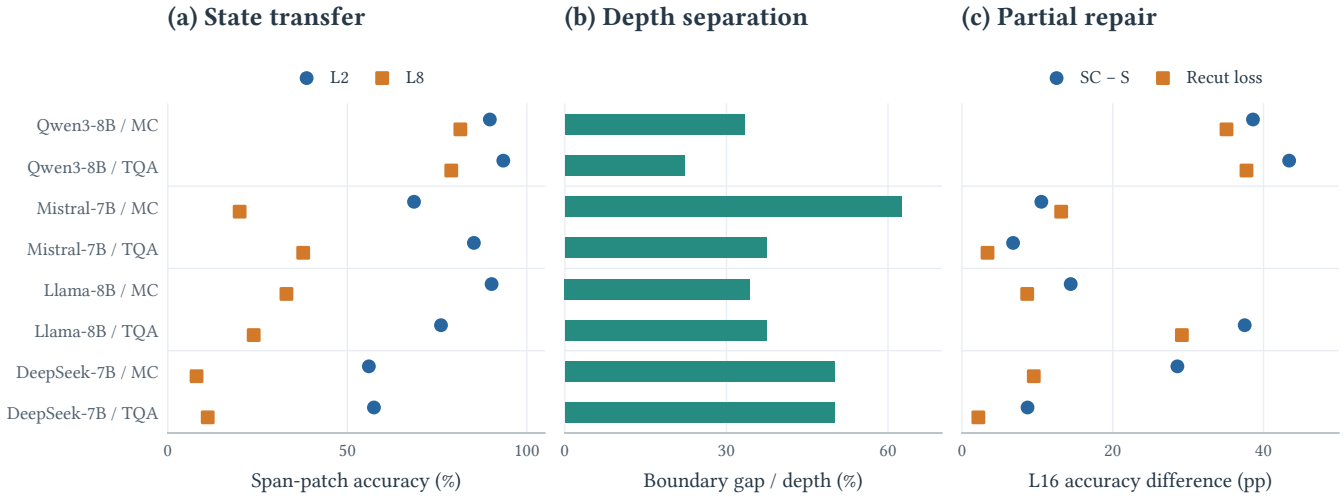


Figure 1: Overview of all four main models on both benchmarks (MC: MedCalc; TQA: TheoremQA). (a) Skill-span replacement accuracy at layers 2 and 8, with each battery's own baseline-screened population. (b) The gap between first transfer and dependence boundaries, as a percentage of model depth, on common eligible items. (c) Fixed-layer-16 joint-repair gain over state replacement and the loss after a next-layer recut. Panels use separate protocol-specific populations, not a single item-level trajectory.

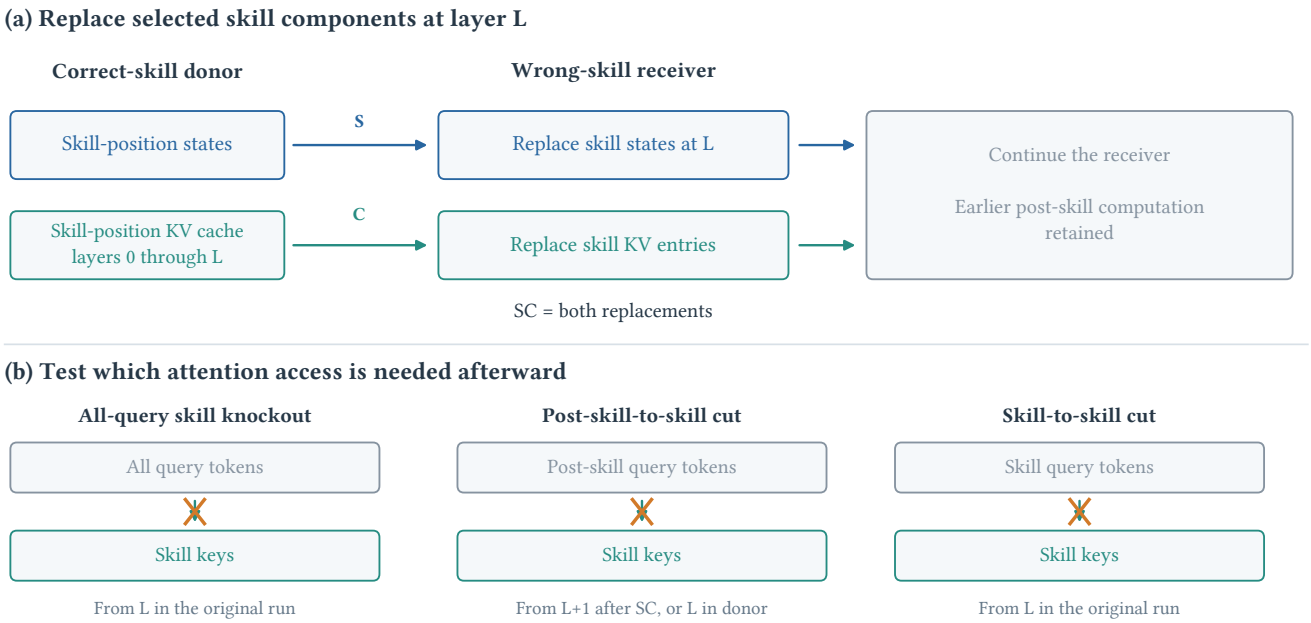


Figure 2: Intervention design. (a) The correct-skill donor supplies skill residual states (S), skill-position key/value (KV) entries through layer L (C), or both (SC) to the wrong-skill receiver. Earlier computation at post-skill positions is retained. (b) Attention cuts distinguish access by post-skill tokens from updates within the skill text. After SC, the post-skill-to-skill cut begins at $L+1$. In the original correct-skill run, path cuts begin at L. All cuts continue through the remaining prompt processing and decoding.

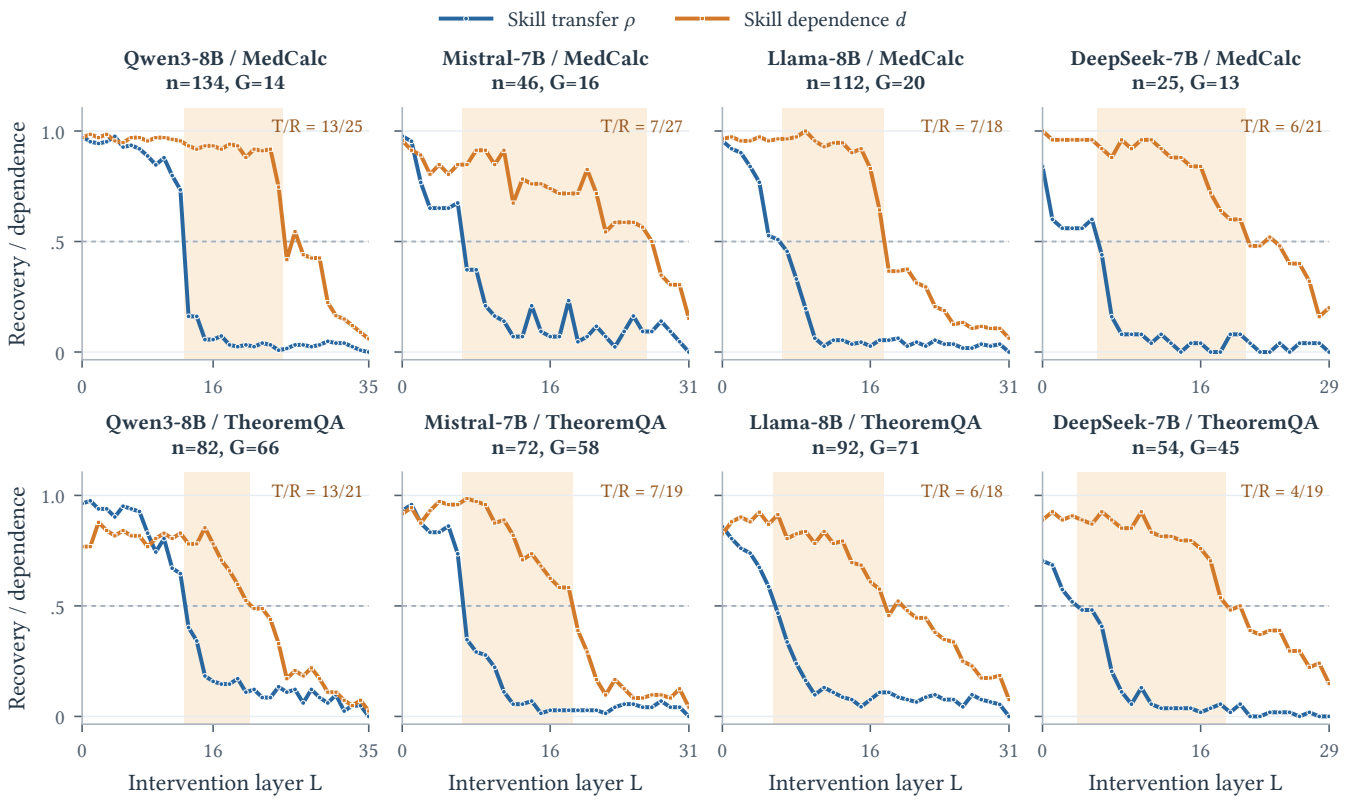


Figure 3: Depth profiles for all eight main model–benchmark comparisons. Blue shows skill-span transfer recovery $\rho(L)$ in a wrong-skill receiver. Orange shows aggregate skill-attention dependence $d(L) = 1 - q(L)$: the fraction of originally correct answers lost after cumulative attention knockout in the correct-skill execution. Shading spans the first half-effect boundaries, highlighting weak transfer despite strong dependence; later recrossings remain visible. Boundary layers are marked above each panel. Each panel uses common eligible item IDs for both curves, with each run's own baselines and execution protocol; G denotes skill groups. The knockout affects both access from post-skill tokens and skill-state updates. Every plotted layer is measured.

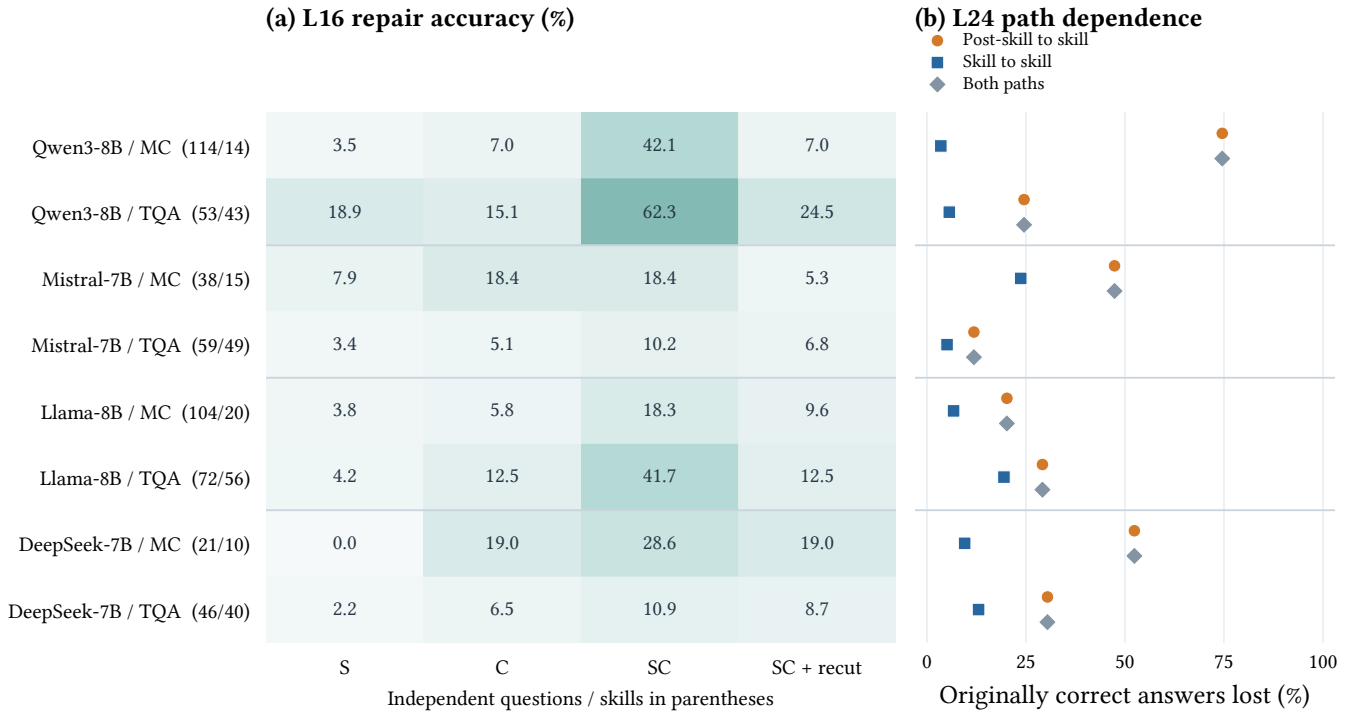


Figure 4: Repair and attention-path results for all eight main combinations. MC denotes MedCalc and TQA denotes TheoremQA. Each row uses its own fresh split-prefill population; parentheses give questions/skill groups. (a) Layer-16 accuracy after state (*S*), cache (*C*), joint (*SC*), or joint replacement followed by a next-layer post-skill-to-skill recut. (b) At layer 24, the fraction of donor-correct answers lost after cutting post-skill-to-skill, skill-to-skill, or both paths. Both prefill and decoding are intervened on. Table 3 gives derived contrasts.

Table 1: State-transfer controls for all eight model–benchmark combinations. Entries are accuracy percentages. n_2/n_8 gives the independent layer-2/layer-8 populations; the remaining controls are at layer 8. Other denotes matched-prefix other-skill content. Qwen3-8B/MedCalc retains its historical layer-8 screen (124/135 correct donor answers); other batteries require fresh correct donors. Layer 8 can already lie beyond a model's transfer boundary.

Model	Task	n_2/n_8	L2 span	L8 span	Last 1	Last 16	Other	Permuted	Noise
Qwen3-8B	MedCalc	97/135	89.7	81.5	3.0	5.9	4.4	4.4	2.2
Qwen3-8B	TheoremQA	61/57	93.4	78.9	10.5	12.3	14.0	22.8	19.3
Mistral-7B	MedCalc	35/35	68.6	20.0	5.7	8.6	2.9	14.3	8.6
Mistral-7B	TheoremQA	61/61	85.2	37.7	1.6	6.6	8.2	14.8	4.9
Llama-3.1-8B	MedCalc	112/112	90.2	33.0	2.7	2.7	8.0	4.5	4.5
Llama-3.1-8B	TheoremQA	92/92	76.1	23.9	12.0	4.3	9.8	14.1	9.8
DeepSeek-7B	MedCalc	25/25	56.0	8.0	0.0	4.0	8.0	4.0	4.0
DeepSeek-7B	TheoremQA	54/54	57.4	11.1	1.9	11.1	5.6	5.6	0.0

Table 2: Main depth comparison across four open-weight models and two benchmarks. n/G gives common eligible questions/skill groups; T and R are the first half-effect crossings for transfer ($\rho < 0.5$) and aggregate skill dependence ($d \leq 0.5$). $\Delta = R - T$ is in layers; Δ/D normalizes the gap by the model's layer count. All eight cells are measured on their own baseline-screened, common-item populations.

Model	MedCalc-Bench					TheoremQA				
	n/G	T	R	Δ	Δ/D (%)	n/G	T	R	Δ	Δ/D (%)
Qwen3-8B	134/14	13	25	12	33.3	82/66	13	21	8	22.2
Mistral-7B	46/16	7	27	20	62.5	72/58	7	19	12	37.5
Llama-3.1-8B	112/20	7	18	11	34.4	92/71	6	18	12	37.5
DeepSeek-7B	25/13	6	21	15	50.0	54/45	4	19	15	50.0

Table 3: Mechanism contrasts for all eight main combinations, in percentage points. n/G gives each row's questions/skills. With accuracy a_X, interaction is $a_{SC} - a_S - a_C + a_R$; recut loss is $a_{SC} - a_{SC+\text{recut}}$. The L24 path gap is skill-to-skill-cut accuracy minus post-skill-to-skill-cut accuracy, equivalently the excess loss from cutting post-skill access.

Model	Task	n/G	L16: $SC - S$	L16: interaction	L16: recut loss	L24: path gap
Qwen3-8B	MedCalc	114/14	+38.6	+31.6	+35.1	+71.1
Qwen3-8B	TheoremQA	53/43	+43.4	+28.3	+37.7	+18.9
Mistral-7B	MedCalc	38/15	+10.5	−7.9	+13.2	+23.7
Mistral-7B	TheoremQA	59/49	+6.8	+1.7	+3.4	+6.8
Llama-3.1-8B	MedCalc	104/20	+14.4	+8.7	+8.7	+13.5
Llama-3.1-8B	TheoremQA	72/56	+37.5	+25.0	+29.2	+9.7
DeepSeek-7B	MedCalc	21/10	+28.6	+9.5	+9.5	+42.9
DeepSeek-7B	TheoremQA	46/40	+8.7	+2.2	+2.2	+17.4

Table 4: Cross-question transfer and repeated-write controls for all eight main combinations. Current/Other compare skill-after-question donors at layer 8; $n_{\text{cross}}/n_{\text{win}}$ gives independent populations. W indicates the inclusive write window. Acc. is raw accuracy; the historical Qwen3-8B/MedCalc row uses normalized recovery ρ with its own donor/receiver baselines. Rows with different units are not accuracy rankings. Windows ending at the last layer depend on model depth.

Model	Task	$n_{\text{cross}}/n_{\text{win}}$	Unit	Current	Other	W8–11	W12–15	W14–19	W16–end
Qwen3-8B	MedCalc	151/135	ρ	1.02	0.98	0.89	0.74	0.16	0.06
Qwen3-8B	TheoremQA	8/61	Acc.	0.88	0.88	0.79	0.61	0.38	0.25
Mistral-7B	MedCalc	10/35	Acc.	0.40	0.10	0.20	0.14	0.09	0.09
Mistral-7B	TheoremQA	7/61	Acc.	0.86	0.57	0.38	0.07	0.07	0.07
Llama-3.1-8B	MedCalc	92/112	Acc.	0.50	0.39	0.33	0.05	0.04	0.03
Llama-3.1-8B	TheoremQA	46/92	Acc.	0.41	0.33	0.24	0.11	0.08	0.08
DeepSeek-7B	MedCalc	25/25	Acc.	0.96	0.92	0.08	0.08	0.00	0.04
DeepSeek-7B	TheoremQA	23/54	Acc.	0.39	0.35	0.11	0.04	0.04	0.02

Table 5: State/cache repair at the fixed layer 16 on all main model–task combinations. Accuracies are percentages. Gain is $a_{SC} - a_S$; recut loss is $a_{SC} - a_{SC+\text{recut}}$, both in percentage points. Each row uses its own n questions from G skill groups.

Model	Task	n/G	S	C	SC	SC+recut	Gain (pp)	Recut loss (pp)
Qwen3-8B	MedCalc	114/14	3.5	7.0	42.1	7.0	38.6	35.1
Qwen3-8B	TheoremQA	53/43	18.9	15.1	62.3	24.5	43.4	37.7
Mistral-7B	MedCalc	38/15	7.9	18.4	18.4	5.3	10.5	13.2
Mistral-7B	TheoremQA	59/49	3.4	5.1	10.2	6.8	6.8	3.4
Llama-3.1-8B	MedCalc	104/20	3.8	5.8	18.3	9.6	14.4	8.7
Llama-3.1-8B	TheoremQA	72/56	4.2	12.5	41.7	12.5	37.5	29.2
DeepSeek-7B	MedCalc	21/10	0.0	19.0	28.6	19.0	28.6	9.5
DeepSeek-7B	TheoremQA	46/40	2.2	6.5	10.9	8.7	8.7	2.2

Table 6: Independent eligible question counts for the four main models. Cross-question counts require an available other-question donor. The Qwen3-8B/MedCalc layer-8 battery retains its historical group screen; other rows apply their own unpatched baselines. Within a battery, variable-donor controls can have smaller populations.

Model	Task	L8 battery	L2 battery	Cross-question	Windows	Repair	Path cuts
Qwen3-8B	MedCalc	135	97	151	135	114	114
Qwen3-8B	TheoremQA	57	61	8	61	53	53
Mistral-7B	MedCalc	35	35	10	35	38	38
Mistral-7B	TheoremQA	61	61	7	61	59	59
Llama-3.1-8B	MedCalc	112	112	92	112	104	104
Llama-3.1-8B	TheoremQA	92	92	46	92	72	72
DeepSeek-7B	MedCalc	25	25	25	25	21	21
DeepSeek-7B	TheoremQA	54	54	23	54	46	46